arXiv:2310.08864

Open X-Embodiment: Robotic Learning Datasets and RT-X Models

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Abstract

Large, high-capacity models trained on diverse datasets have shown remarkable successes on efficiently tackling downstream applications. We argue that to enable general robotic learning policies, we should train generalist X-embodiment models. We assemble a dataset from 22 different robots collected through a collaboration between 21 institutions, demonstrating 527 skills (160266 tasks). We show that policies trained across many embodiments and varied environments exhibit positive transfer.

Introduction

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Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2310.08864, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for Open X-Embodiment: Robotic Learning Datasets and RT-X Models follows the abstract: Large, high-capacity models trained on diverse datasets have shown remarkable successes on efficiently tackling downstream applications. We argue that to enable general robotic learning policies, we should train generalist X-embodiment models. We assemble a dataset from 22 different robots collected through a collaboration between 21 institutions, demonstrating 527 skills (160266 tasks). We show that policies trained across many embodiments and varied environments exhibit positive transfer. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

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Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: Large, high-capacity models trained on diverse datasets have shown remarkable successes on efficiently tackling downstream applications. We argue that to enable general robotic learning policies, we should train generalist X-embodiment models. We assemble a dataset from 22 different robots collected through a collaboration between 21 institutions, demonstrating

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Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

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We conclude by summarizing the contribution of arXiv:2310.08864 and outline directions for follow-up work.

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References

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